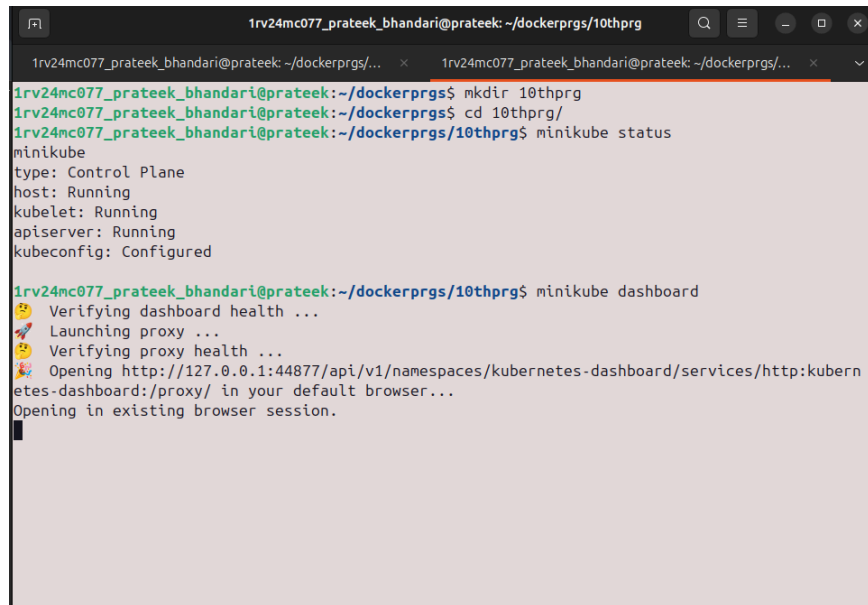


Program 10

Step 1: Start Minikube with resources

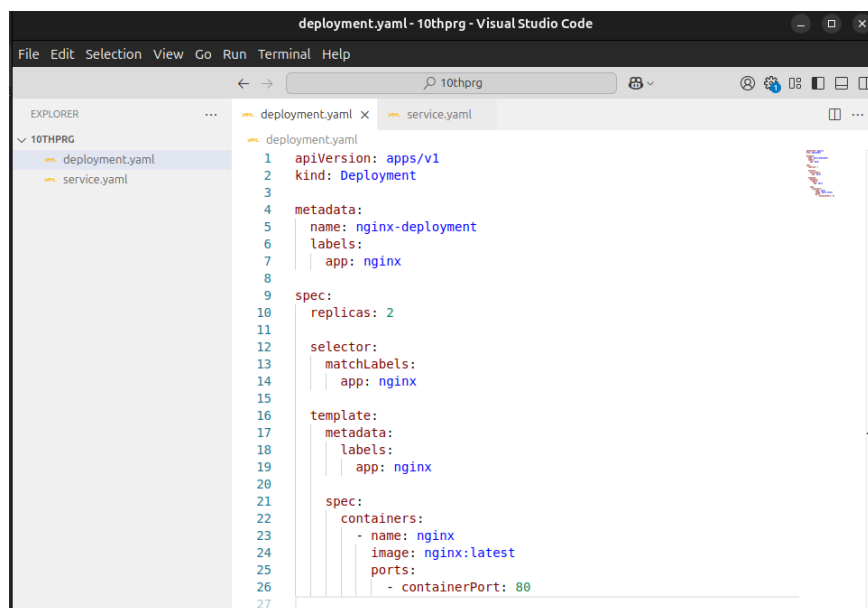
Step 2: Open Kubernetes Dashboard



```
1rv24mc077_prateek_bhandari@prateek: ~/dockerprgs/10thprg
1rv24mc077_prateek_bhandari@prateek:~/dockerprgs$ mkdir 10thprg
1rv24mc077_prateek_bhandari@prateek:~/dockerprgs$ cd 10thprg/
1rv24mc077_prateek_bhandari@prateek:~/dockerprgs/10thprg$ minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured

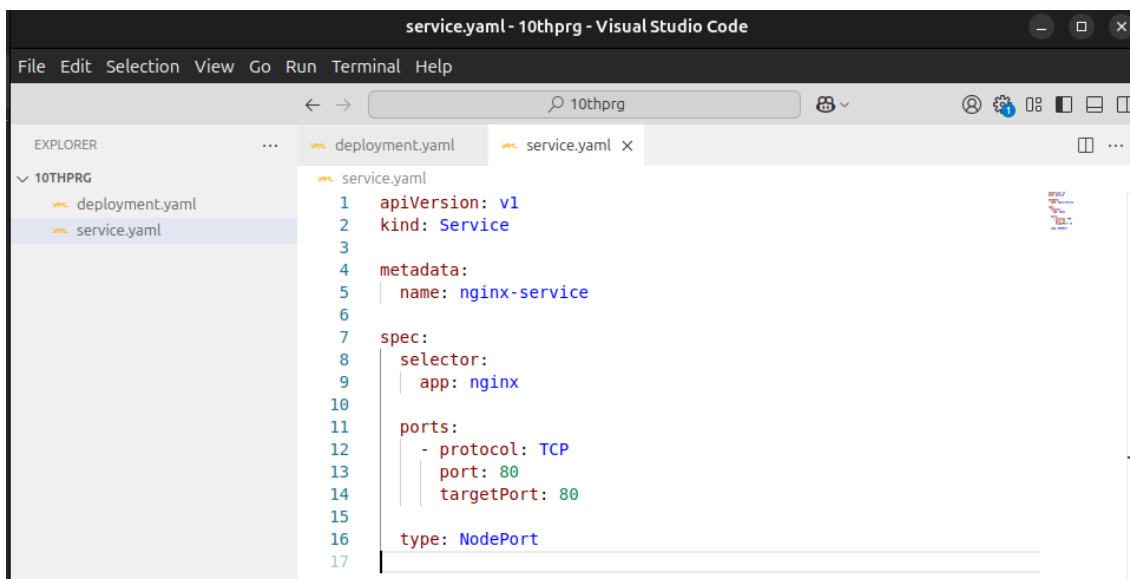
1rv24mc077_prateek_bhandari@prateek:~/dockerprgs/10thprg$ minikube dashboard
🐳 Verifying dashboard health ...
🔌 Launching proxy ...
🐳 Verifying proxy health ...
🔗 Opening http://127.0.0.1:44877/api/v1/namespaces/kubernetes-dashboard/services/http:kubern
etes-dashboard:/proxy/ in your default browser...
🔗 Opening in existing browser session.
```

Step 3: create a Deployment file → deployment.yaml



```
deployment.yaml - 10thprg - Visual Studio Code
File Edit Selection View Go Run Terminal Help
10thprg
EXPLORER
  10THPRG
    deployment.yaml
    service.yaml
deployment.yaml
1  apiVersion: apps/v1
2  kind: Deployment
3
4  metadata:
5    name: nginx-deployment
6    labels:
7      app: nginx
8
9  spec:
10   replicas: 2
11
12   selector:
13     matchLabels:
14       app: nginx
15
16   template:
17     metadata:
18       labels:
19         app: nginx
20
21     spec:
22       containers:
23         - name: nginx
24           image: nginx:latest
25           ports:
26             - containerPort: 80
27
```

Step 4: Create a Service file → service.yaml



The screenshot shows the Visual Studio Code editor with the file 'service.yaml' open. The Explorer sidebar on the left shows a project named '10THPRG' containing 'deployment.yaml' and 'service.yaml'. The 'service.yaml' file is selected and its content is displayed in the editor. The content is a Kubernetes Service definition for 'nginx-service'.

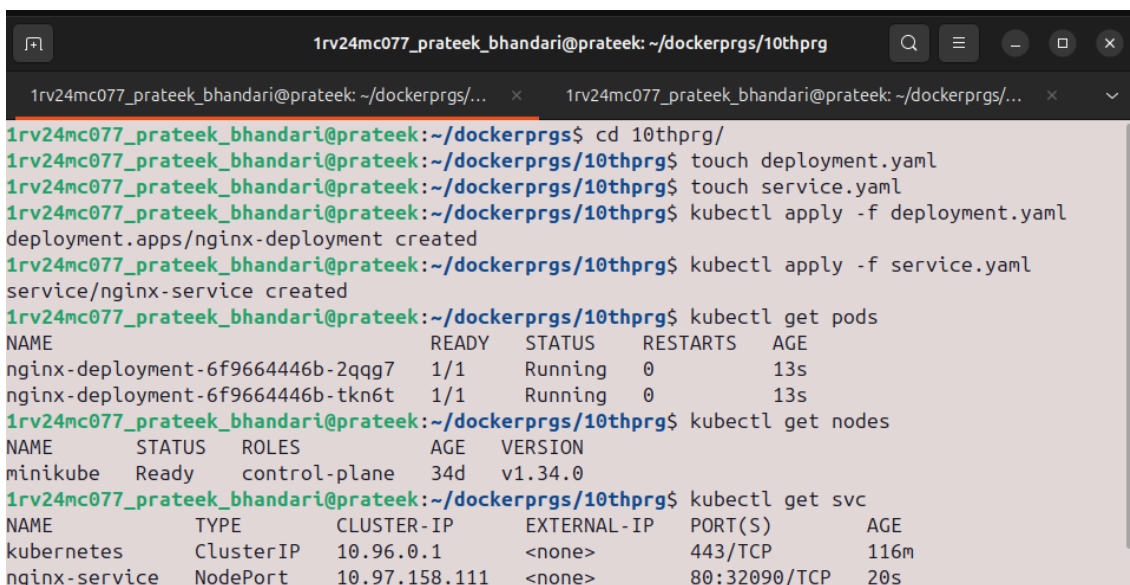
```
1 apiVersion: v1
2 kind: Service
3
4 metadata:
5   name: nginx-service
6
7 spec:
8   selector:
9     app: nginx
10
11   ports:
12     - protocol: TCP
13       port: 80
14       targetPort: 80
15
16   type: NodePort
17
```

Step 5: Apply Resources

Step 6: Monitor via CLI

Step 7: Check get nodes

Step 8: Check get services



The screenshot shows a terminal window with the following commands and output:

```
1rv24mc077_prateek_bhandari@prateek: ~/dockerprgs/10thprg
1rv24mc077_prateek_bhandari@prateek:~/dockerprgs$ cd 10thprg/
1rv24mc077_prateek_bhandari@prateek:~/dockerprgs/10thprg$ touch deployment.yaml
1rv24mc077_prateek_bhandari@prateek:~/dockerprgs/10thprg$ touch service.yaml
1rv24mc077_prateek_bhandari@prateek:~/dockerprgs/10thprg$ kubectl apply -f deployment.yaml
deployment.apps/nginx-deployment created
1rv24mc077_prateek_bhandari@prateek:~/dockerprgs/10thprg$ kubectl apply -f service.yaml
service/nginx-service created
1rv24mc077_prateek_bhandari@prateek:~/dockerprgs/10thprg$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-6f9664446b-2qqg7   1/1     Running   0           13s
nginx-deployment-6f9664446b-tnk6t   1/1     Running   0           13s
1rv24mc077_prateek_bhandari@prateek:~/dockerprgs/10thprg$ kubectl get nodes
NAME      STATUS   ROLES    AGE   VERSION
minikube  Ready    control-plane  34d   v1.34.0
1rv24mc077_prateek_bhandari@prateek:~/dockerprgs/10thprg$ kubectl get svc
NAME            TYPE       CLUSTER-IP   EXTERNAL-IP   PORT(S)          AGE
kubernetes      ClusterIP  10.96.0.1    <none>        443/TCP          116m
nginx-service    NodePort   10.97.158.111 <none>        80:32090/TCP     20s
```

Step 9: Check pod logs

Step 10: Describe Pods

```
1rv24mc077_prateek_bhandari@prateek:~/dockerprgs/10thprg$ kubectl logs nginx-deployment-6f9664446b-2qqg7
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/01/04 13:14:24 [notice] 1#1: using the "epoll" event method
2026/01/04 13:14:24 [notice] 1#1: nginx/1.29.4
2026/01/04 13:14:24 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/01/04 13:14:24 [notice] 1#1: OS: Linux 6.14.0-37-generic
2026/01/04 13:14:24 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2026/01/04 13:14:24 [notice] 1#1: start worker processes
2026/01/04 13:14:24 [notice] 1#1: start worker process 28
2026/01/04 13:14:24 [notice] 1#1: start worker process 29
2026/01/04 13:14:24 [notice] 1#1: start worker process 30
2026/01/04 13:14:24 [notice] 1#1: start worker process 31
1rv24mc077_prateek_bhandari@prateek:~/dockerprgs/10thprg$ kubectl describe nginx-deployment-6f9664446b-2qqg7
error: the server doesn't have a resource type "nginx-deployment-6f9664446b-2qqg7"
1rv24mc077_prateek_bhandari@prateek:~/dockerprgs/10thprg$ kubectl describe pod nginx-deployment-6f9664446b-2qqg7

Name:          nginx-deployment-6f9664446b-2qqg7
Namespace:     default
Priority:       0
Service Account: default
Node:          minikube/192.168.49.2
```

Step 11: access the application via Minikube & Viewing the same in the web browser

